

Zeta-Law Entropy, Gamma Curvature, and Laplace-Measure Applications

Pudim AI Project

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Abstract

This staged paper presents a unified public theory built around Laplace-transform positivity: zeta-law entropy is reorganized by logarithmic compression, gamma-ratio and polygamma inequalities are expressed through completely monotone kernels, and new applications are obtained by transporting measure positivity through Stieltjes, complete-Bernstein, stable-subordination, and generalized- Stieltjes operations. Compared with version v008, this version adds ten fully solved applications. They include a Stieltjes representation for a Ramanujan integral, a stable-subordination representation of a Prabhakar measure, a derivative-level Mills-ratio bound family, two negative answers to over-strong Bernstein and complete-monotonic-degree conjectures, exact threshold theorems for tau, W-symbol, and digamma-cutoff problems, a proof of a Gurland logarithmic-complete-monotonicity conjecture, and a triangular compression formula for Sokal's generalized-Stieltjes expressions. The application ledger now records twenty-three public applications, all with uniform status *Solved*.

Keywords. Riemann zeta function; zeta probability law; entropy; Laplace transform; complete monotonicity; Bernstein functions; Stieltjes functions; gamma function; digamma function; Mills ratio; generalized Stieltjes functions.

1 Global notation and positivity conventions

The Riemann zeta function is denoted by ζ , and Euler's gamma function is denoted by Γ . The digamma function is $\psi = \Gamma'/\Gamma$, its higher derivatives are the polygamma functions $\psi^{(m)}$, and Euler's constant is $\gamma = -\psi(1)$. We write $\mathbb{N} = \{1, 2, \dots\}$, $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, and use $\mathbb{E}_\mu$, $\mathbb{P}_\mu$, and Var_μ for expectation, probability, and variance under a measure μ . When the subscript is β , these symbols refer to the zeta law $\rho_\beta(n) = n^{-\beta}/\zeta(\beta)$ on $\mathbb{N}$, later recorded formally in Definition 4.1. The rising factorial, or Pochhammer symbol, is $(a)_m = a(a+1) \cdots (a+m-1)$ for $m \geq 1$, with $(a)_0 = 1$. The notation $[z^m]F(z)$ means coefficient extraction from the power series F .

A C^∞ function $f : (0, \infty) \rightarrow \mathbb{R}$ is completely monotone if

$$(-1)^m f^{(m)}(x) \geq 0 \quad (m \in \mathbb{N}_0, x > 0),$$

and it is strictly completely monotone if all these inequalities are strict whenever the derivative is not identically zero. A positive function f is logarithmically completely monotone in this paper if $\log f$ has the same alternating derivative sign pattern on $(0, \infty)$. A nonnegative C^∞ function B is a Bernstein function if B' is completely monotone. A complete Bernstein function is a Bernstein function with the standard Stieltjes-density representation; equivalently, in the applications below,

it is obtained by integrating a completely monotone density against $1 - e^{-xt}$. A Stieltjes function is a function representable in the form

$$f(x) = a + \frac{b}{x} + \int_0^\infty \frac{1}{x+t} d\mu(t), \quad a, b \geq 0,$$

with μ a positive measure satisfying the usual local integrability condition. A generalized Stieltjes function of order $\lambda > 0$ is a positive superposition of kernels $(x+t)^{-\lambda}$, in the sense used by Sokal. Source-specific special functions and measures are introduced locally before the theorem that uses them.

2 Motivation and main results

Many classical identities for $\zeta(s)$ become more transparent after a probabilistic normalization. For $\beta > 1$, the weights $n^{-\beta}$ define a Gibbs law whose partition function is precisely $\zeta(\beta)$. The energy is $\log n$, the free energy is $\log \zeta(\beta)$, and ratios such as $\zeta(\beta+r)/\zeta(\beta)$ become moments. This viewpoint is elementary, but it packages several unrelated-looking zeta inequalities into a common calculus.

The paper develops six connected layers. The moment layer records basic zeta-law identities. The modular entropy layer compares the successor map $n \mapsto n+1$ with its finite residue shadows modulo q . The Euler-score layer rewrites zeta energy through von Mangoldt divisibility events. The Mellin-Planck layer treats $\Gamma(s)\zeta(s)$ as a continuous partition function and converts generalized Holder inequalities into integral norm inequalities. The tail layer studies $\zeta_n(s)^{-1}$ as a reciprocal partition problem for the truncated zeta law. The v009 Laplace-transport layer then carries the same positivity language into Stieltjes, complete-Bernstein, stable-subordination, determinant, and generalized-Stieltjes applications.

Section 3 records twenty-three solved external problems with stable application labels. APP-0001 through APP-0013 are the earlier zeta-law, Gamma, digamma, tail, beta-window, and q-log-convexity entries. APP-0014 through APP-0023 form the new v009 block: a Ramanujan integral Stieltjes classification, a Prabhakar stable-subordination formula, a Mills-ratio derivative hierarchy, two counterexamples to over-strong Bernstein or degree claims, exact tau, W-symbol, and digamma cutoffs, a Gurland log-complete-monotonicity proof, and a Sokal triangular compression formula. These results are presented through a common measure-transport mechanism.

3 Applications

The public application ledger contains twenty-three solved entries. The first thirteen were staged in earlier theory versions. The v009 layer adds ten more entries, all unconditional and all reducible to the same structural principle: find the positive kernel, then transport it through an operation that preserves complete monotonicity or one of its Stieltjes/Bernstein descendants.

1. **APP-0001.** Alzer–Kwong convexity and concavity problem. Solved by Theorem 9.2; source problem reference [3].
2. **APP-0002.** Nantomah zeta positivity problem. Solved by Theorem 9.1; source problem reference [1].
3. **APP-0003.** Sroysang generalized Holder problem. Solved by Theorem 9.3; source problem reference [2].

4. **APP-0004.** Complete monotonicity of $(\log \Gamma(x) + \log \Gamma(1/x))''$. Solved by Theorem 8.5; source problem reference [4].
5. **APP-0005.** Exact inverse-tail floor formula at $s = 7$. Solved by Theorem 5.2; source context [6, 7].
6. **APP-0006.** Exact inverse-tail floor formula at $s = 8$. Solved by Theorem 5.3; source context [6, 7].
7. **APP-0007.** Complete monotonicity of $-(\psi(x)\psi(1/x))''$. Solved by Theorem 4.4; source problem reference [4].
8. **APP-0008.** Counterexample to complete monotonicity of higher-order polygamma product curvature. Solved by Theorem 4.5; source problem reference [4].
9. **APP-0009.** Sharp reciprocal Gamma-product monotonicity threshold. Solved by Theorem 8.4; source problem reference [5].
10. **APP-0010.** Nielsen k -beta derivative-ratio monotonicity. Solved by Theorem 10.2; source problem reference [8].
11. **APP-0011.** Exact $n = 2$ beta-window endpoint. Solved by Theorem 11.1; source problem reference [4].
12. **APP-0012.** Baricz V_q strict q -log-convexity. Solved by Theorem 11.2; source problem reference [9].
13. **APP-0013.** Yin (p, k) -digamma sharp α -necessity. Solved by Theorem 11.3; source problem reference [10].
14. **APP-0014.** Ramanujan integral Stieltjes/complete-Bernstein classification. Solved by Theorem 12.1; source problem reference [11].
15. **APP-0015.** Prabhakar Q-measure stable-subordination representation. Solved by Theorem 12.2; source problem reference [12].
16. **APP-0016.** Mills-ratio all-derivative-level bound extractor. Solved by Theorem 12.3; source problem reference [13].
17. **APP-0017.** Gamma-quotient Bernstein-for-all counterexample. Solved by Theorem 12.4; source problem reference [14].
18. **APP-0018.** Qi-Guo tau-threshold supremum correction. Solved by Theorem 12.5; source problem reference [15].
19. **APP-0019.** Wakrim W-symbol exact Bernstein range. Solved by Theorem 12.6; source problem reference [16].
20. **APP-0020.** Qi h_λ degree-four conjecture refutation. Solved by Theorem 12.7; source problem reference [17].
21. **APP-0021.** Szabo digamma-cutoff theorem. Solved by Theorem 12.8; source problem reference [18].

22. **APP-0022.** Chen–Choi Gurland gamma-ratio logarithmic complete monotonicity. Solved by Theorem 12.9; source problem reference [19].
23. **APP-0023.** Sokal generalized-Stieltjes lambda-derivative compression. Solved by Theorem 12.10; source problem reference [20].

4 Zeta-law calculus

Definition 4.1 (Riemann zeta probability law). For $\beta > 1$, define a probability law ρ_β on $\mathbb{N}$ by

$$\rho_\beta(n) = \frac{n^{-\beta}}{\zeta(\beta)}, \quad n \in \mathbb{N}. \quad (4.1)$$

Equivalently, with $E(n) = \log n$ and $Z(\beta) = \zeta(\beta)$,

$$\rho_\beta(n) = \frac{e^{-\beta E(n)}}{Z(\beta)}. \quad (4.2)$$

Definition 4.2 (Zeta free energy). The zeta free energy is

$$A(\beta) = \log \zeta(\beta), \quad \beta > 1. \quad (4.3)$$

Proposition 4.3 (Free-energy derivatives). *For every $\beta > 1$,*

$$A'(\beta) = -\mathbb{E}_\beta[\log N], \quad A''(\beta) = \text{Var}_\beta(\log N). \quad (4.4)$$

Proof. Absolute convergence of the differentiated Dirichlet series on compact subintervals of $(1, \infty)$ permits termwise differentiation. Thus

$$A'(\beta) = \frac{\zeta'(\beta)}{\zeta(\beta)} = -\frac{\sum_{n \geq 1} (\log n) n^{-\beta}}{\zeta(\beta)} = -\mathbb{E}_\beta[\log N]. \quad (4.5)$$

Differentiating once more gives

$$A''(\beta) = \mathbb{E}_\beta[(\log N)^2] - \mathbb{E}_\beta[\log N]^2 = \text{Var}_\beta(\log N). \quad (4.6)$$

□

Qi, Lim, and Nantomah ask whether $P_0(x) = \psi(x)\psi(1/x)$ is concave, and more strongly whether $-P_0''$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers the stronger question.

Theorem 4.4 (APP-0007: Complete monotonicity of reciprocal digamma product curvature). *Let*

$$P_0(x) = \psi(x)\psi(1/x), \quad x > 0. \quad (4.7)$$

Then $-P_0''$ is strictly completely monotonic on $(0, \infty)$. In particular, P_0 is strictly concave on $(0, \infty)$.

Proof. Direct differentiation gives

$$\begin{aligned} P_0''(x) &= \psi''(x)\psi(1/x) - \frac{2}{x^2}\psi'(x)\psi'(1/x) \\ &\quad + \frac{2}{x^3}\psi(x)\psi'(1/x) + \frac{1}{x^4}\psi(x)\psi''(1/x). \end{aligned} \quad (4.8)$$

Use the shifted Weierstrass expansion

$$\psi(x) = -\gamma - \frac{1}{x} + \sum_{m=1}^{\infty} a_m(x), \quad a_m(x) = \frac{x}{m(m+x)}, \quad (4.9)$$

and, after substituting $1/x$,

$$\psi(1/x) = -\gamma - x + \sum_{n=1}^{\infty} b_n(x), \quad b_n(x) = \frac{1}{n(nx+1)}. \quad (4.10)$$

On compact subintervals of $(0, \infty)$, the series for a_m , b_n , and their first two derivatives are uniformly summable; the differentiated double products are dominated by products of summable majorants. Hence P_0 may be differentiated term by term twice.

Let $A_0(x) = -\gamma - 1/x$ and $B_0(x) = -\gamma - x$. The base term satisfies

$$-\frac{d^2}{dx^2}(A_0(x)B_0(x)) = -\frac{2\gamma}{x^3}. \quad (4.11)$$

For $n \geq 1$, elementary rational differentiation and Laplace inversion give

$$-\frac{d^2}{dx^2}(A_0(x)b_n(x)) = \int_0^{\infty} e^{-xt} t^2 \frac{n + (\gamma - n)e^{-t/n}}{n^2} dt, \quad (4.12)$$

and, for $m \geq 1$,

$$-\frac{d^2}{dx^2}(B_0(x)a_m(x)) = \int_0^{\infty} e^{-xt} t^2 (m - \gamma) e^{-mt} dt. \quad (4.13)$$

For $mn > 1$,

$$-\frac{d^2}{dx^2}(a_m(x)b_n(x)) = \int_0^{\infty} e^{-xt} t^2 \left[\frac{e^{-t/n}}{mn^2(mn-1)} - \frac{e^{-mt}}{n(mn-1)} \right] dt. \quad (4.14)$$

The exceptional product is $a_1(x)b_1(x) = x/(x+1)^2$, and

$$-\frac{d^2}{dx^2} \frac{x}{(x+1)^2} = \int_0^{\infty} e^{-xt} t^2 (t-1) e^{-t} dt. \quad (4.15)$$

The identities above follow from

$$\frac{1}{(x+c)^3} = \int_0^{\infty} e^{-xt} \frac{t^2}{2} e^{-ct} dt, \quad \frac{1}{(x+c)^4} = \int_0^{\infty} e^{-xt} \frac{t^3}{6} e^{-ct} dt. \quad (4.16)$$

The coefficients of the double products group by the two identities

$$\sum_{m=1}^{\infty} \frac{1}{m(mn-1)} = -\psi(1-1/n) - \gamma \quad (n \geq 2) \quad (4.17)$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n(mn-1)} = -\psi(1-1/m) - \gamma \quad (m \geq 2). \quad (4.18)$$

Both are immediate from $\psi(z+1) + \gamma = \sum_{k=1}^{\infty} z/(k(k+z))$, with $z = -1/n$ and $z = -1/m$. Combining (4.11)–(4.18) gives

$$-P_0''(x) = \int_0^{\infty} e^{-xt} K(t) dt, \quad (4.19)$$

where

$$K(t) = t^2 \left[1 - \gamma + (t-1)e^{-t} + \sum_{n=2}^{\infty} \frac{n(1 - e^{-t/n}) - \psi(1 - 1/n)e^{-t/n}}{n^2} + \sum_{m=2}^{\infty} (m + \psi(1 - 1/m))e^{-mt} \right]. \quad (4.20)$$

It remains to prove $K(t) > 0$. Since ψ is increasing and $1 - 1/n \in [1/2, 1)$,

$$-\psi(1 - 1/n) > 0 \quad (n \geq 2). \quad (4.21)$$

Also

$$m + \psi(1 - 1/m) \geq 2 - \gamma - 2 \log 2 > 0 \quad (m \geq 2), \quad (4.22)$$

because $\psi(1 - 1/m) \geq \psi(1/2) = -\gamma - 2 \log 2$. Thus the two sums in (4.20) are positive term by term, apart from the possible need to dominate $1 - \gamma + (t-1)e^{-t}$.

For $n \geq 2$,

$$-\psi(1 - 1/n) = \gamma + \int_{1-1/n}^1 \psi'(u) du > \gamma + \int_{1-1/n}^1 \frac{du}{u^2} = \gamma + \frac{1}{n-1}. \quad (4.23)$$

Consequently the n -sum in (4.20) is bounded below by

$$Ae^{-t/2}, \quad A = \gamma(\zeta(2) - 1) + 2 - \zeta(2). \quad (4.24)$$

Therefore $K(t)/t^2$ is bounded below by

$$f(t) = 1 - \gamma + (t-1)e^{-t} + Ae^{-t/2}. \quad (4.25)$$

Now

$$f(0) = A - \gamma = (2 - \zeta(2))(1 - \gamma) > 0, \quad \lim_{t \rightarrow \infty} f(t) = 1 - \gamma > 0. \quad (4.26)$$

Moreover,

$$f'(t) = e^{-t} \left(2 - t - \frac{A}{2} e^{t/2} \right), \quad (4.27)$$

and the factor in parentheses is strictly decreasing. Hence f has at most one critical point, and any such point is a maximum. Its minimum on $[0, \infty)$ is attained at an endpoint, so $f(t) > 0$. Thus $K(t) > 0$ for $t > 0$.

Differentiating under (4.19) yields

$$(-1)^r \frac{d^r}{dx^r} (-P_0''(x)) = \int_0^{\infty} t^r e^{-xt} K(t) dt > 0 \quad (4.28)$$

for all $r \geq 0$. This proves strict complete monotonicity of $-P_0''$, and the case $r = 0$ gives strict concavity of P_0 . $\square$

Qi, Lim, and Nantomah also ask whether the higher-order products $P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x)$, $n \in \mathbb{N}$, are convex, and more strongly whether P_n'' is completely monotonic. The following result refutes the stronger assertion. It does not settle the weaker convexity question.

Theorem 4.5 (APP-0008: Counterexample to complete monotonicity of higher-order polygamma product curvature). *Let*

$$P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x), \quad n \in \mathbb{N}, \quad x > 0. \quad (4.29)$$

Then P_n'' is not completely monotonic for all $n \in \mathbb{N}$. More precisely, for every $n \geq 29$,

$$P_n'''(2) > 0, \quad (4.30)$$

and, already at lower order,

$$P_7^{(6)}(3) < 0. \quad (4.31)$$

Proof. For $m \geq 1$, define the sign-normalized polygamma function

$$A_m(x) = (-1)^{m+1}\psi^{(m)}(x) = m! \sum_{k=0}^{\infty} (x+k)^{-m-1}. \quad (4.32)$$

Then $A_m(x) > 0$, $A'_m(x) = -A_{m+1}(x)$, and $P_n(x) = A_n(x)A_n(1/x)$. Differentiating twice gives

$$\begin{aligned} P_n''(x) = & A_{n+2}(x)A_n(1/x) - \frac{2}{x^2}A_{n+1}(x)A_{n+1}(1/x) \\ & - \frac{2}{x^3}A_n(x)A_{n+1}(1/x) + \frac{1}{x^4}A_n(x)A_{n+2}(1/x). \end{aligned} \quad (4.33)$$

First prove (4.30). Put $p = n + 1$, and normalize

$$Q_p(x) = \frac{P_n(x)}{(n!)^2} = \sum_{m, \ell \geq 0} \left(\frac{x}{(x+m)(1+\ell x)} \right)^p. \quad (4.34)$$

At $x = 2$, the term $(m, \ell) = (0, 0)$ is constant, and the unique largest nonconstant base is $(m, \ell) = (1, 0)$, equal to $2/3$. Write

$$Q_p(x) = 1 + \left(\frac{x}{x+1} \right)^p + R_p(x). \quad (4.35)$$

For $h(x) = x/(x+1)$, direct differentiation gives

$$\frac{d^3}{dx^3} h(x)^p = p \left(\frac{x}{x+1} \right)^p \frac{p^2 - 6px - 3p + 6x^2 + 6x + 2}{x^3(x+1)^3}. \quad (4.36)$$

Therefore

$$\left. \frac{d^3}{dx^3} h(x)^p \right|_{x=2} = \frac{p(p^2 - 15p + 38)}{216} \left(\frac{2}{3} \right)^p. \quad (4.37)$$

For $p \geq 30$, this is at least

$$\frac{p^3}{432} \left(\frac{2}{3} \right)^p. \quad (4.38)$$

For a general summand

$$F_{m, \ell}(x) = \left(\frac{x}{(x+m)(1+\ell x)} \right)^p, \quad (4.39)$$

let $u_{m,\ell} = d(\log F_{m,\ell}^{1/p})/dx$. At $x = 2$,

$$|u_{m,\ell}| \leq \frac{1}{2}, \quad |u'_{m,\ell}| \leq \frac{1}{4}, \quad |u''_{m,\ell}| \leq \frac{1}{4}. \quad (4.40)$$

Since

$$F'''_{m,\ell} = F_{m,\ell} (p^3 u_{m,\ell}^3 + 3p^2 u_{m,\ell} u'_{m,\ell} + p u''_{m,\ell}), \quad (4.41)$$

we have $|F'''_{m,\ell}(2)| \leq p^3 F_{m,\ell}(2)$ for $p \geq 2$. The remaining mass in R_p is bounded by

$$\sum_{(m,\ell) \neq (0,0),(1,0)} F_{m,\ell}(2) < 8 \cdot 2^{-p} \quad (p \geq 3). \quad (4.42)$$

Indeed, the $m \geq 2, \ell = 0$ part is at most $3 \cdot 2^{-p}$, while the $\ell \geq 1$ part is at most $2 \cdot 2^{-p}(1 + (2/3)^p + 3 \cdot 2^{-p})$. Combining (4.38) and (4.42),

$$Q_p'''(2) \geq p^3 2^{-p} \left[\frac{1}{432} \left(\frac{4}{3} \right)^p - 8 \right]. \quad (4.43)$$

Because $(4/3)^{30} > 432 \cdot 8$, (4.43) is positive for $p \geq 30$. Hence $P_n'''(2) = (n!)^2 Q_{n+1}'''(2) > 0$ for $n \geq 29$. Complete monotonicity of P_n'' would require $(P_n'')' = P_n''' \leq 0$, so P_n'' is not completely monotonic for $n \geq 29$.

We now record the lower-order obstruction (4.31). For bookkeeping, put

$$T_{p,i,j}(x) = x^{-p} A_i(x) A_j(1/x). \quad (4.44)$$

The identity $A'_m = -A_{m+1}$ gives the exact recurrence

$$\frac{d}{dx} T_{p,i,j}(x) = -p T_{p+1,i,j}(x) - T_{p,i+1,j}(x) + T_{p+2,i,j+1}(x). \quad (4.45)$$

Starting with (4.33), applying (4.45) four times with $n = 7$ expresses $P_7^{(6)}(3)$ as an exact finite linear combination of 22 terms $T_{p,i,j}(3)$.

The sign of this finite expression can be certified by rational intervals. For rational $x > 0$ and $N \geq 1$, monotone integral comparison in (4.32) gives

$$m! \left(\sum_{k=0}^{N-1} (x+k)^{-m-1} + \frac{(x+N)^{-m}}{m} \right) \leq A_m(x), \quad (4.46)$$

and

$$A_m(x) \leq m! \left(\sum_{k=0}^{N-1} (x+k)^{-m-1} + \frac{(x+N)^{-m}}{m} + (x+N)^{-m-1} \right). \quad (4.47)$$

Using $N = 80$ in (4.46) and (4.47) for every $A_m(3)$ and $A_m(1/3)$ appearing in the 22-term expression, and choosing upper or lower endpoints according to the sign of each coefficient, yields

$$-74998.154649422399828421756528409981771664439555587 < P_7^{(6)}(3), \quad (4.48)$$

and

$$P_7^{(6)}(3) < -74997.331312439140215056900814096498575289999496021. \quad (4.49)$$

Thus $P_7^{(6)}(3) < 0$. If P_7'' were completely monotonic, then

$$(-1)^4 \frac{d^4}{dx^4} P_7''(3) = P_7^{(6)}(3) \geq 0, \quad (4.50)$$

contradicting (4.49). Hence the proposed complete-monotonicity strengthening for all $n \in \mathbb{N}$ is false in both an infinite high-order family and already at $n = 7$. $\square$

Proposition 4.6 (Zeta-law calculus). *Let $\beta > 1$, and let N have law ρ_β . For every $r \geq 0$,*

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \quad (4.51)$$

Moreover, for $\alpha, \beta > 1$,

$$D(\rho_\alpha \| \rho_\beta) = (\beta - \alpha) \mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha). \quad (4.52)$$

Proof. The moment identity follows by direct summation:

$$\mathbb{E}_\beta[N^{-r}] = \sum_{n=1}^{\infty} n^{-r} \frac{n^{-\beta}}{\zeta(\beta)} = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \quad (4.53)$$

For relative entropy, substitute (4.1):

$$D(\rho_\alpha \| \rho_\beta) = \sum_{n=1}^{\infty} \rho_\alpha(n) ((\beta - \alpha) \log n + \log \zeta(\beta) - \log \zeta(\alpha)), \quad (4.54)$$

which is (4.52). □

5 Tail zeta laws and inverse-tail floors

Definition 5.1 (Tail zeta partition function). For $s > 1$ and $n \in \mathbb{N}$, define the tail zeta partition function by

$$Z_{s,n} = \zeta_n(s) = \sum_{k=n}^{\infty} k^{-s}. \quad (5.1)$$

The corresponding tail zeta law is the probability distribution on $\{n, n+1, \dots\}$ given by

$$\rho_{s,n}(k) = \frac{k^{-s}}{\zeta_n(s)}, \quad k \geq n. \quad (5.2)$$

Exact formulas for integer parts of reciprocal zeta tails have been studied for small integer exponents. Known formulas include $s = 2, 3, 4, 5$, with a sufficiently-large- n formula for $s = 6$; Kim and Song record that no analogous formula was known for integer $s > 6$ at the time of their work [6]. The 2024 asymptotic inverse-tail results of Pan and Wu provide related context [7]. The next two theorems give exact formulas at $s = 7$ and $s = 8$. The first uses a single rational telescoping enclosure, while the second needs adjacent corrected asymptotic truncations.

Theorem 5.2 (APP-0005: Exact inverse-tail floor formula at $s=7$). *Let*

$$T_7(n) = \zeta_n(7) = \sum_{k=n}^{\infty} \frac{1}{k^7}. \quad (5.3)$$

Define

$$Q(n) = 120n^6 - 360n^5 + 660n^4 - 720n^3 + 354n^2 - 54n + 375, \quad P(n) = \frac{Q(n)}{20}. \quad (5.4)$$

Then, for every $n \geq 28$,

$$\lfloor T_7(n)^{-1} \rfloor = \lfloor P(n) \rfloor. \quad (5.5)$$

For $1 \leq n \leq 27$, the exact values are

n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$
1	0	10	4495760	19	241765529
2	119	11	8167814	20	331399044
3	1862	12	14061542	21	447175152
4	12573	13	23143975	22	594869693
5	54069	14	36668777	23	781167220
6	175597	15	56228085	24	1013751716
7	471118	16	83808666	25	1301401638
8	1100904	17	121852400	26	1654089273
9	2317459	18	173321080	27	2083084422

(5.6)

Proof. The polynomial Q is positive on $\mathbb{N}$, because

$$Q(n) = 120(n-1)^6 + 360(n-1)^5 + 660(n-1)^4 + 720(n-1)^3 + 354(n-1)^2 + 54(n-1) + 375. \quad (5.7)$$

For $k \geq 1$, direct expansion gives

$$\frac{1}{k^7} - \left(\frac{20}{Q(k)} - \frac{20}{Q(k+1)} \right) = \frac{9(60284k^4 + 29176k^2 + 15625)}{k^7 Q(k) Q(k+1)} > 0. \quad (5.8)$$

Summing (5.8) over $k \geq n$ yields

$$T_7(n) > \frac{20}{Q(n)} = \frac{1}{P(n)}, \quad (5.9)$$

and therefore $T_7(n)^{-1} < P(n)$.

For $k \geq 28$, another expansion gives

$$\left(\frac{20}{Q(k) - 3} - \frac{20}{Q(k+1) - 3} \right) - \frac{1}{k^7} = \frac{36(20k^6 - 14961k^4 - 7235k^2 - 3844)}{k^7(Q(k) - 3)(Q(k+1) - 3)}. \quad (5.10)$$

The numerator in (5.10) is positive for $k \geq 28$. Indeed, writing $k = m + 28$, it becomes

$$20m^6 + 3360m^5 + 220239m^4 + 7105168m^3 + 114013021m^2 + 751143512m + 436261580, \quad (5.11)$$

which has positive coefficients for $m \geq 0$. Hence, for $n \geq 28$,

$$T_7(n) < \frac{20}{Q(n) - 3} = \frac{1}{P(n) - 3/20}. \quad (5.12)$$

Combining (5.9) and (5.12) gives

$$P(n) - \frac{3}{20} < T_7(n)^{-1} < P(n). \quad (5.13)$$

Modulo 20,

$$Q(n) \equiv 14n^2 - 14n + 15 \pmod{20}. \quad (5.14)$$

Since $n(n-1) \pmod{10} \in \{0, 2, 6\}$, this implies

$$Q(n) \pmod{20} \in \{3, 15, 19\}. \quad (5.15)$$

Thus the fractional part of $P(n)$ is one of $3/20, 15/20, 19/20$. If $a = \lfloor P(n) \rfloor$, then $P(n) - 3/20 \geq a$, and (5.13) gives

$$a < T_7(n)^{-1} < a + 1. \quad (5.16)$$

This proves (5.5) for $n \geq 28$.

For the finite range, $T_7(1) > 1$, so $0 < T_7(1)^{-1} < 1$. For $2 \leq n \leq 27$, let M_n denote the corresponding value in (5.6). With $N = 1500$, exact rational arithmetic verifies

$$\sum_{k=n}^N \frac{1}{k^7} > \frac{1}{M_n + 1} \quad (5.17)$$

and

$$\sum_{k=n}^N \frac{1}{k^7} + \frac{1}{6N^6} < \frac{1}{M_n}. \quad (5.18)$$

The monotone integral estimate

$$\sum_{k=N+1}^{\infty} \frac{1}{k^7} < \int_N^{\infty} x^{-7} dx = \frac{1}{6N^6} \quad (5.19)$$

then gives

$$\frac{1}{M_n + 1} < T_7(n) < \frac{1}{M_n}. \quad (5.20)$$

Therefore $M_n < T_7(n)^{-1} < M_n + 1$, proving (5.6). The finite rational certificate is stored with the source as a reproducible calculation. $\square$

Theorem 5.3 (APP-0006: Exact inverse-tail floor formula at $s=8$). *Let*

$$T_8(n) = \zeta_n(8) = \sum_{k=n}^{\infty} \frac{1}{k^8}. \quad (5.21)$$

Define

$$\begin{aligned} A_8(n) = & 7n^7 - \frac{49}{2}n^6 + \frac{637}{12}n^5 - \frac{1715}{24}n^4 + \frac{7399}{144}n^3 - \frac{1715}{96}n^2 + \frac{69685}{1728}n - \frac{65611}{3456} \\ & - \frac{23764069}{103680n} - \frac{23764069}{207360n^2} + \frac{2102131409}{1244160n^3} + \frac{2149659547}{829440n^4} \\ & - \frac{203290800013}{14929920n^5} - \frac{1146003910541}{29859840n^6} + \frac{21587525169497}{179159040n^7}. \end{aligned} \quad (5.22)$$

Then, for every $n \geq 6$,

$$\lfloor T_8(n)^{-1} \rfloor = \lfloor A_8(n) \rfloor. \quad (5.23)$$

For $1 \leq n \leq 5$, the exact values are

$$\frac{n}{\lfloor T_8(n)^{-1} \rfloor} \left| \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 0 & 245 & 5844 & 53503 & 291407 \end{array} \right|. \quad (5.24)$$

Proof. Set

$$B_8(n) = A_8(n) + \frac{200335112892761}{358318080n^8}. \quad (5.25)$$

For $C = A_8$ or B_8 , write $R_C(k) = 1/C(k)$. Exact denominator clearing gives the two telescoping inequalities

$$R_{A_8}(k) - R_{A_8}(k+1) - \frac{1}{k^8} > 0 \quad (k \geq 3) \quad (5.26)$$

and

$$\frac{1}{k^8} - (R_{B_8}(k) - R_{B_8}(k+1)) > 0 \quad (k \geq 1). \quad (5.27)$$

In (5.26), the cleared numerator is a polynomial in $k - 3$ with all coefficients positive; in (5.27), the cleared numerator is a polynomial in $k - 1$ with all coefficients positive. The exact coefficient verification is recorded in the rational certificate accompanying the manuscript.

Summing (5.26) over $k \geq n$ gives

$$T_8(n) < R_{A_8}(n) = \frac{1}{A_8(n)} \quad (5.28)$$

for $n \geq 3$, and hence $A_8(n) < T_8(n)^{-1}$. Summing (5.27) gives

$$T_8(n) > R_{B_8}(n) = \frac{1}{B_8(n)}, \quad (5.29)$$

and hence $T_8(n)^{-1} < B_8(n)$. Therefore, for $n \geq 6$,

$$A_8(n) < T_8(n)^{-1} < B_8(n). \quad (5.30)$$

It remains to see that no integer lies between $A_8(n)$ and $B_8(n)$. Exact integer arithmetic verifies

$$\lfloor A_8(n) \rfloor = \lfloor B_8(n) \rfloor \quad (6 \leq n \leq 10^6). \quad (5.31)$$

For $n > 10^6$, let $P_8(n)$ be the polynomial part of $A_8(n)$, through the constant term. The numerator of $P_8(n)$ modulo 3456 runs through the odd residue classes, so $P_8(n)$ is at distance at least $1/3456$ from the next integer. The correction $A_8(n) - P_8(n)$ is negative and has absolute value $< 1/3456$, while

$$0 < B_8(n) - A_8(n) < \frac{1}{3456}. \quad (5.32)$$

Thus $B_8(n)$ is still below the next integer after $A_8(n)$. Combining this with (5.30) proves (5.23).

For $1 \leq n \leq 5$, (5.24) is certified by exact rational partial sums. With $N = 300$, the displayed value M_n satisfies, for $2 \leq n \leq 5$,

$$\sum_{k=n}^N \frac{1}{k^8} > \frac{1}{M_n + 1} \quad (5.33)$$

and

$$\sum_{k=n}^N \frac{1}{k^8} + \frac{1}{7N^7} < \frac{1}{M_n}. \quad (5.34)$$

Since

$$\sum_{k=N+1}^{\infty} \frac{1}{k^8} < \int_N^{\infty} x^{-8} dx = \frac{1}{7N^7}, \quad (5.35)$$

we get $M_n < T_8(n)^{-1} < M_n + 1$. For $n = 1$, $T_8(1) > 1$, so the floor is 0. $\square$

6 Modular successor entropy

Definition 6.1 (Modular residue distribution and successor entropy). For $q \geq 2$, define the residue distribution of ρ_β modulo q by

$$\mu_{q,\beta}(a) = \sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n), \quad a \in \mathbb{Z}/q\mathbb{Z}. \quad (6.1)$$

Writing $a + 1$ cyclically in $\mathbb{Z}/q\mathbb{Z}$, define

$$B_q(\beta) = \sum_{a \in \mathbb{Z}/q\mathbb{Z}} \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\mu_{q,\beta}(a+1)}. \quad (6.2)$$

Theorem 6.2 (Zeta-law successor entropy and modular resolution). *For every $\beta > 1$, define*

$$\Sigma(\beta) = \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (6.3)$$

Then

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta+k), \quad (6.4)$$

and

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \rightarrow \infty} B_q(\beta). \quad (6.5)$$

Proof. Since

$$\frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \left(1 + \frac{1}{n}\right)^\beta, \quad (6.6)$$

we have

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{n=1}^{\infty} n^{-\beta} \log \left(1 + \frac{1}{n}\right). \quad (6.7)$$

Using

$$\log(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k \quad (0 < x \leq 1) \quad (6.8)$$

with Abel limiting at $x = 1$, and summing against $n^{-\beta}$, gives (6.4).

For the modular identity, fix $q \geq 2$. The log-sum inequality gives, for each residue class a ,

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} \geq \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\sum_{n \equiv a \pmod{q}} \rho_\beta(n+1)}. \quad (6.9)$$

For $a \neq 0$, the denominator is $\mu_{q,\beta}(a+1)$. For $a = 0$, it is $\mu_{q,\beta}(1) - \rho_\beta(1) \leq \mu_{q,\beta}(1)$. Summing (6.9) over a gives

$$\Sigma(\beta) \geq B_q(\beta) \quad (q \geq 2). \quad (6.10)$$

Conversely, fix $M \in \mathbb{N}$. If $q > M + 1$, then for $1 \leq a \leq M + 1$,

$$\mu_{q,\beta}(a) = \rho_\beta(a) + O_{\beta,M}(q^{-\beta}), \quad (6.11)$$

and $\mu_{q,\beta}(0) = q^{-\beta}$. All cyclic terms in $B_q(\beta)$ are nonnegative except possibly the boundary term, which satisfies

$$\mu_{q,\beta}(0) \log \frac{\mu_{q,\beta}(0)}{\mu_{q,\beta}(1)} = O_\beta(q^{-\beta} \log q). \quad (6.12)$$

Therefore

$$\liminf_{q \rightarrow \infty} B_q(\beta) \geq \sum_{n=1}^M \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (6.13)$$

Letting $M \rightarrow \infty$ gives $\liminf_q B_q(\beta) \geq \Sigma(\beta)$, which together with (6.10) proves (6.5). $\square$

Corollary 6.3 (Prime-modulus Dirichlet L-resolution). *Let p be prime, let χ range over Dirichlet characters modulo p , and define*

$$S_a(p, \beta) = \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi) \quad (1 \leq a \leq p-1). \quad (6.14)$$

Then

$$\mu_{p,\beta}(a) = \frac{S_a(p, \beta)}{(p-1)\zeta(\beta)} \quad (1 \leq a \leq p-1), \quad \mu_{p,\beta}(0) = p^{-\beta}. \quad (6.15)$$

Consequently the prime-modulus functionals obtained by substituting (6.15) into $B_p(\beta)$ satisfy

$$\Sigma(\beta) = \lim_{\substack{p \rightarrow \infty \\ p \text{ prime}}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}) = \sup_{p \text{ prime}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}). \quad (6.16)$$

Proof. For $a \not\equiv 0 \pmod{p}$, character orthogonality gives

$$\frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} \chi(n) = \begin{cases} 1, & n \equiv a \pmod{p}, \\ 0, & n \not\equiv a \pmod{p}. \end{cases} \quad (6.17)$$

Therefore

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{p}}} n^{-\beta} = \frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi). \quad (6.18)$$

Dividing by $\zeta(\beta)$ gives the nonzero residue formula, while

$$\mu_{p,\beta}(0) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (mp)^{-\beta} = p^{-\beta}. \quad (6.19)$$

The functional identity follows by substituting these residue formulas into $B_p(\beta)$ and applying Theorem 6.2 along the prime sequence. $\square$

7 Euler-score identities

Theorem 7.1 (Euler-score decomposition). *For every $\beta > 1$,*

$$-(\log \zeta)'(\beta) = \mathbb{E}_\beta[\log N] = \sum_{d=1}^{\infty} \Lambda(d) \mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}. \quad (7.1)$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}. \quad (7.2)$$

Proof. The von Mangoldt identity is

$$\log n = \sum_{d|n} \Lambda(d). \quad (7.3)$$

By nonnegative summation,

$$\mathbb{E}_\beta[\log N] = \sum_{n \geq 1} \rho_\beta(n) \sum_{d|n} \Lambda(d) = \sum_{d \geq 1} \Lambda(d) \mathbb{P}_\beta(d | N). \quad (7.4)$$

The divisibility probability is

$$\mathbb{P}_\beta(d | N) = \frac{1}{\zeta(\beta)} \sum_{m \geq 1} (dm)^{-\beta} = d^{-\beta}. \quad (7.5)$$

Substitution into (7.4) proves (7.1); (7.2) follows from (4.5). $\square$

Proposition 7.2 (Finite Euler-score identity). *For every $A \subseteq \{1, \dots, N\}$,*

$$\sum_{n \in A} \log n = \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md). \quad (7.6)$$

Proof. Reverse the order of summation:

$$\sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md) = \sum_{n \in A} \sum_{d|n} \Lambda(d) = \sum_{n \in A} \log n. \quad (7.7)$$

$\square$

8 Curvature and Mellin-Planck tools

Lemma 8.1 (Uniform positive-axis curvature bound). *For every $s \geq 3$,*

$$(\log \zeta)''(s) = \text{Var}_s(\log N) < \frac{1}{3}. \quad (8.1)$$

Proof. By Proposition 4.3,

$$(\log \zeta)''(s) = \frac{\zeta''(s)}{\zeta(s)} - \left(\frac{\zeta'(s)}{\zeta(s)} \right)^2 \leq \frac{\zeta''(s)}{\zeta(s)} \leq \zeta''(s). \quad (8.2)$$

For $s \geq 3$,

$$\zeta''(s) = \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^s} \leq \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3}. \quad (8.3)$$

The function $x \mapsto (\log x)^2 x^{-3}$ is decreasing for $x \geq 2$, so

$$\sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3} \leq \frac{(\log 2)^2}{8} + \int_2^{\infty} \frac{(\log x)^2}{x^3} dx. \quad (8.4)$$

Since

$$\int_2^{\infty} \frac{(\log x)^2}{x^3} dx = \frac{(\log 2)^2 + \log 2 + \frac{1}{2}}{8}, \quad (8.5)$$

we obtain

$$(\log \zeta)''(s) \leq \frac{2(\log 2)^2 + \log 2 + \frac{1}{2}}{8} < \frac{1}{3}. \quad (8.6)$$

$\square$

Definition 8.2 (Mellin-Planck partition function). For $s > 1$, define

$$M(s) = \Gamma(s)\zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt. \quad (8.7)$$

The corresponding probability law is

$$\nu_s(dt) = \frac{t^{s-1}}{(e^t - 1)M(s)} dt. \quad (8.8)$$

Proposition 8.3 (Mellin-Planck log-convexity). For $s > 1$,

$$\frac{d^2}{ds^2} \log M(s) = \text{Var}_{\nu_s}(\log t) \geq 0. \quad (8.9)$$

Proof. Differentiate the logarithm of the integral in (8.7) under the integral sign. The first derivative is the ν_s -expectation of $\log t$, and the second derivative is its variance:

$$\frac{d^2}{ds^2} \log M(s) = \mathbb{E}_{\nu_s}[(\log t)^2] - \mathbb{E}_{\nu_s}[\log t]^2. \quad (8.10)$$

□

Bulboacă and Zayed proved several monotonicity criteria for functions of the form $u(s)/(\Gamma(s + \rho)\Gamma(s))$ and asked for best possible positive values of ρ in their examples [5]. The next application gives the sharp endpoint in the base reciprocal Gamma-product case and the corresponding endpoint criterion for monotone logarithmic derivatives.

Theorem 8.4 (APP-0009: Sharp reciprocal Gamma-product monotonicity threshold). Let $\psi = \Gamma'/\Gamma$, let γ be Euler's constant, and let $\rho_* > 0$ be the unique solution of

$$\psi(1 + \rho_*) = \gamma. \quad (8.11)$$

For $\rho > -1$, define

$$\varphi_\rho(s) = \frac{1}{\Gamma(s + \rho)\Gamma(s)}, \quad s \geq 1. \quad (8.12)$$

Then φ_ρ is strictly decreasing on $[1, \infty)$ if and only if

$$\rho \geq \rho_*. \quad (8.13)$$

Equivalently,

$$W_\rho(s) = \Gamma(s + \rho)\Gamma(s) \quad (8.14)$$

is strictly increasing on $[1, \infty)$ if and only if (8.13) holds. Numerically,

$$\rho_* = 1.2583969670859318174106234224981693941 \dots \quad (8.15)$$

More generally, let $u : [1, \infty) \rightarrow (0, \infty)$ be differentiable and suppose

$$J(s) = \frac{u'(s)}{u(s)} \quad (8.16)$$

is nonincreasing on $[1, \infty)$. Then

$$\Phi_{\rho, u}(s) = \frac{u(s)}{\Gamma(s + \rho)\Gamma(s)} \quad (8.17)$$

is strictly decreasing on $[1, \infty)$ if and only if

$$\psi(1 + \rho) \geq \gamma + J(1). \quad (8.18)$$

Proof. The digamma function is strictly increasing on $(0, \infty)$, since

$$\psi'(x) = \sum_{k=0}^{\infty} \frac{1}{(x+k)^2} > 0. \quad (8.19)$$

Hence (8.11) has a unique positive solution.

For $s \geq 1$, the logarithmic derivative of (8.14) is

$$\frac{d}{ds} \log W_{\rho}(s) = \psi(s + \rho) + \psi(s). \quad (8.20)$$

The right-hand side is strictly increasing in s . Its minimum on $[1, \infty)$ is therefore

$$\psi(1 + \rho) + \psi(1) = \psi(1 + \rho) - \gamma. \quad (8.21)$$

If $\rho \geq \rho_*$, then (8.21) is nonnegative, and it is positive for every $s > 1$. Thus, for $1 \leq a < b$,

$$\log W_{\rho}(b) - \log W_{\rho}(a) = \int_a^b (\psi(s + \rho) + \psi(s)) ds > 0. \quad (8.22)$$

So W_{ρ} is strictly increasing and $\varphi_{\rho} = 1/W_{\rho}$ is strictly decreasing. Conversely, if $\rho < \rho_*$, then (8.21) is negative, so by continuity W_{ρ} decreases initially and φ_{ρ} increases initially. This proves the base equivalence.

For the general statement,

$$\frac{d}{ds} \log \Phi_{\rho,u}(s) = J(s) - \psi(s + \rho) - \psi(s). \quad (8.23)$$

The term J is nonincreasing by hypothesis, while $\psi(s + \rho) + \psi(s)$ is strictly increasing. Therefore the logarithmic derivative in (8.23) is strictly decreasing, and its maximum on $[1, \infty)$ is

$$J(1) - \psi(1 + \rho) - \psi(1) = J(1) - \psi(1 + \rho) + \gamma. \quad (8.24)$$

The function $\Phi_{\rho,u}$ is strictly decreasing exactly when this maximum is nonpositive, which is (8.18). If the inequality fails, the logarithmic derivative is positive near $s = 1$, so monotonicity fails. $\square$

Qi, Lim, and Nantomah asked whether the second derivative of $\log \Gamma(x) + \log \Gamma(1/x)$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers this in the affirmative by a reciprocal-Weierstrass Laplace kernel.

Theorem 8.5 (APP-0004: Complete monotonicity of reciprocal-Gamma curvature). *Let*

$$H(x) = \log \Gamma(x) + \log \Gamma(1/x), \quad x > 0. \quad (8.25)$$

Then H'' is strictly completely monotonic on $(0, \infty)$; that is,

$$(-1)^m \frac{d^m}{dx^m} H''(x) > 0 \quad (m \geq 0, x > 0). \quad (8.26)$$

Proof. The Weierstrass product gives, for $x > 0$,

$$\log \Gamma(x) = -\log x - \gamma x - \sum_{n=1}^{\infty} \left(\log \left(1 + \frac{x}{n} \right) - \frac{x}{n} \right). \quad (8.27)$$

After substituting $1/x$ in (8.27), adding the two formulas, and differentiating twice on compact subintervals of $(0, \infty)$, the logarithmic second-derivative terms cancel and

$$H''(x) = \sum_{n=1}^{\infty} \frac{1}{(x+n)^2} - \frac{2\gamma}{x^3} + \sum_{n=1}^{\infty} \left[\frac{1}{(x+1/n)^2} - \frac{1}{x^2} + \frac{2}{nx^3} \right]. \quad (8.28)$$

The elementary Laplace identities

$$\frac{1}{(x+a)^2} = \int_0^{\infty} t e^{-(x+a)t} dt, \quad \frac{1}{x^2} = \int_0^{\infty} t e^{-xt} dt, \quad \frac{2a}{x^3} = a \int_0^{\infty} t^2 e^{-xt} dt \quad (8.29)$$

show that, with $a = 1/n$,

$$\frac{1}{(x+1/n)^2} - \frac{1}{x^2} + \frac{2}{nx^3} = \int_0^{\infty} e^{-xt} t \left(e^{-t/n} - 1 + \frac{t}{n} \right) dt. \quad (8.30)$$

Also,

$$\sum_{n=1}^{\infty} \frac{1}{(x+n)^2} = \int_0^{\infty} e^{-xt} \frac{t}{e^t - 1} dt, \quad (8.31)$$

and

$$-\frac{2\gamma}{x^3} = \int_0^{\infty} e^{-xt} (-\gamma t^2) dt. \quad (8.32)$$

Combining (8.28)–(8.32) gives

$$H''(x) = \int_0^{\infty} e^{-xt} K(t) dt, \quad (8.33)$$

where

$$K(t) = \frac{t}{e^t - 1} + t \sum_{n=1}^{\infty} \left(e^{-t/n} - 1 + \frac{t}{n} \right) - \gamma t^2. \quad (8.34)$$

It remains to prove $K(t) > 0$. Put

$$\phi(u) = e^{-u} - 1 + u. \quad (8.35)$$

Then $\phi \geq 0$, $\phi' = 1 - e^{-u} \geq 0$, and $u \mapsto \phi(t/u)$ is positive and decreasing on $[1, \infty)$. Therefore

$$\sum_{n=1}^{\infty} \phi(t/n) \geq \int_1^{\infty} \phi(t/u) du. \quad (8.36)$$

With $v = t/u$, integration by parts gives

$$\int_1^{\infty} \phi(t/u) du = t \int_0^t \frac{\phi(v)}{v^2} dv = -\phi(t) + t \int_0^t \frac{1 - e^{-v}}{v} dv. \quad (8.37)$$

Using

$$\int_0^t \frac{1 - e^{-v}}{v} dv = \gamma + \log t + E_1(t), \quad E_1(t) = \int_t^{\infty} \frac{e^{-v}}{v} dv, \quad (8.38)$$

we obtain

$$\frac{K(t)}{t} \geq \frac{1}{e^t - 1} - \phi(t) + t \log t + tE_1(t). \quad (8.39)$$

Since $E_1(t) > 0$ and $\log t \geq 1 - 1/t$ for $t > 0$,

$$\frac{K(t)}{t} > \frac{1}{e^t - 1} - \phi(t) + t - 1 = \frac{1}{e^t - 1} - e^{-t} = \frac{e^{-t}}{e^t - 1} > 0. \quad (8.40)$$

Thus $K(t) > 0$ for $t > 0$. Differentiating under the Laplace integral in (8.33) gives

$$(-1)^m \frac{d^m}{dx^m} H''(x) = \int_0^\infty t^m e^{-xt} K(t) dt > 0. \quad (8.41)$$

This proves (8.26). $\square$

9 External zeta inequality proofs

Theorem 9.1 (APP-0002: Affirmative solution of Nantomah zeta positivity problem). *For every $n \in \mathbb{N}$,*

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0. \quad (9.1)$$

Proof. Put $s = n+1 \geq 2$ and $Z_s = \zeta(s)$. The expression in (9.1) becomes

$$K_s = (s+1)Z_s Z_{s+2} - sZ_{s+1}^2 - Z_s Z_{s+1}. \quad (9.2)$$

Let $X(m) = 1/m$ under ρ_s . By (4.51),

$$\mathbb{E}_s[X] = \frac{Z_{s+1}}{Z_s}, \quad \mathbb{E}_s[X^2] = \frac{Z_{s+2}}{Z_s}. \quad (9.3)$$

Thus

$$\frac{K_s}{Z_s^2} = (s+1)\mathbb{E}_s[X^2] - s\mathbb{E}_s[X]^2 - \mathbb{E}_s[X] = (s+1)\text{Var}_s(X) + \mathbb{E}_s[X]^2 - \mathbb{E}_s[X]. \quad (9.4)$$

Write $a = \zeta(s) - 1$, $b = \zeta(s+1) - 1$, and $c = \zeta(s+2) - 1$. Expanding gives

$$K_s = L_s + Q_s, \quad L_s = sa + (s+1)c - (2s+1)b, \quad Q_s = (s+1)ac - sb^2 - ab. \quad (9.5)$$

The linear part is termwise positive:

$$L_s = \sum_{m=2}^\infty m^{-s} \frac{(m-1)(s(m-1)-1)}{m^2} \geq 2^{-s} \frac{s-1}{4}. \quad (9.6)$$

Since $b \leq a/2$ and $c \geq 0$,

$$Q_s \geq -sb^2 - ab \geq -\frac{s+2}{4}a^2. \quad (9.7)$$

For $s \geq 4$,

$$a = \sum_{m=2}^\infty m^{-s} \leq 2^{-s} + \int_2^\infty x^{-s} dx = 2^{-s} \left(1 + \frac{2}{s-1}\right). \quad (9.8)$$

Equations (9.6), (9.7), and (9.8) imply

$$K_s \geq \frac{1}{4} \left[2^{-s}(s-1) - (s+2)2^{-2s} \left(1 + \frac{2}{s-1}\right)^2 \right]. \quad (9.9)$$

It is therefore enough to show

$$2^s(s-1) > (s+2) \left(1 + \frac{2}{s-1}\right)^2. \quad (9.10)$$

Equivalently,

$$F(s) = \frac{2^s(s-1)^3}{(s+2)(s+1)^2} > 1. \quad (9.11)$$

Now $F(4) = 72/25 > 1$, and

$$\frac{d}{ds} \log F(s) = \log 2 + \frac{3}{s-1} - \frac{1}{s+2} - \frac{2}{s+1} > 0 \quad (s \geq 4). \quad (9.12)$$

Thus $K_s > 0$ for $s \geq 4$.

For $s = 2$,

$$K_2 = 3\zeta(2)\zeta(4) - 2\zeta(3)^2 - \zeta(2)\zeta(3). \quad (9.13)$$

Using $\zeta(3) < 5/4$, $\zeta(2) = \pi^2/6$, and $\zeta(4) = \pi^4/90$,

$$K_2 > \frac{\pi^6}{180} - \frac{5\pi^2}{24} - \frac{25}{8} > \frac{10415360504209}{18000000000000} > 0. \quad (9.14)$$

For $s = 3$, the estimates $\zeta(3) > 251/216$, $\zeta(5) > 8051/7776$, and $\zeta(4) < 13/12$ yield

$$K_3 > \frac{251}{216} \left(4 \cdot \frac{8051}{7776} - \frac{13}{12}\right) - 3 \left(\frac{13}{12}\right)^2 = \frac{13783}{419904} > 0. \quad (9.15)$$

Hence $K_s > 0$ for all integers $s \geq 2$, proving the theorem. This answers Nantomah's question [1]. $\square$

Theorem 9.2 (APP-0001: Alzer-Kwong convexity and concavity pattern for reciprocal zeta). *Let $F(x) = 1/\zeta(x)$. For every integer $n \geq 1$,*

$$F''(x) > 0 \quad \text{on } (-4n, -4n+2), \quad (9.16)$$

and

$$F''(x) < 0 \quad \text{on } (-4n-2, -4n). \quad (9.17)$$

Proof. Put $x = -u$, where $u > 2$. The functional equation gives

$$\zeta(-u) = -2^{-u}\pi^{-u-1} \sin\left(\frac{\pi u}{2}\right) \Gamma(u+1)\zeta(u+1), \quad (9.18)$$

and hence

$$\frac{1}{\zeta(-u)} = -\frac{2^u\pi^{u+1}}{\Gamma(u+1)\zeta(u+1)\sin(\pi u/2)}. \quad (9.19)$$

Define the positive function

$$G(u) = \frac{2^u\pi^{u+1}}{\Gamma(u+1)\zeta(u+1)|\sin(\pi u/2)|}. \quad (9.20)$$

On $u \in (4n-2, 4n)$, $\sin(\pi u/2) < 0$, so $F(-u) = G(u)$. On $u \in (4n, 4n+2)$, $\sin(\pi u/2) > 0$, so $F(-u) = -G(u)$. Since second derivatives are unchanged by $x = -u$, it remains to prove $G''(u) > 0$.

Let $\ell(u) = \log G(u)$. Then

$$\ell(u) = u \log 2 + (u+1) \log \pi - \log \Gamma(u+1) - \log \zeta(u+1) - \log \left| \sin \left(\frac{\pi u}{2} \right) \right|. \quad (9.21)$$

Differentiating twice gives

$$\ell''(u) = -\psi'(u+1) - (\log \zeta)''(u+1) + \frac{\pi^2}{4} \csc^2 \left(\frac{\pi u}{2} \right). \quad (9.22)$$

With $s = u+1 > 3$,

$$\psi'(s) = \sum_{k=0}^{\infty} \frac{1}{(s+k)^2} < \int_{s-1}^{\infty} \frac{dt}{t^2} = \frac{1}{s-1} \leq \frac{1}{2}. \quad (9.23)$$

Combining (8.1), (9.23), and $\csc^2(\pi u/2) \geq 1$, we get

$$\ell''(u) > \frac{\pi^2}{4} - \frac{1}{2} - \frac{1}{3} = \frac{\pi^2}{4} - \frac{5}{6} > 0. \quad (9.24)$$

Since $G > 0$,

$$G''(u) = G(u)(\ell''(u) + \ell'(u)^2) > 0. \quad (9.25)$$

The two signs in (9.16) and (9.17) now follow from $F(-u) = G(u)$ on $u \in (4n-2, 4n)$ and $F(-u) = -G(u)$ on $u \in (4n, 4n+2)$. This proves the Alzer–Kwong sign pattern [3]. $\square$

Theorem 9.3 (APP-0003: Generalized Holder inequality for Gamma zeta). *Let $m \geq 2$, $r \geq 1$, $x_1, \dots, x_m > 1$, and $p_1, \dots, p_m > 1$ satisfy*

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}. \quad (9.26)$$

Define

$$T = r \sum_{i=1}^m \frac{x_i}{p_i}. \quad (9.27)$$

Then $T > 1$ and

$$\zeta(T) \leq \frac{\prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}}{\Gamma(T)}. \quad (9.28)$$

Proof. Set

$$\lambda_i = \frac{r}{p_i}. \quad (9.29)$$

Then $\lambda_i > 0$, $\sum_i \lambda_i = 1$, and $T = \sum_i \lambda_i x_i > 1$. Define

$$q_i = \frac{p_i}{r}. \quad (9.30)$$

Then $\sum_i 1/q_i = 1$, and the assumptions imply $q_i > 1$ for all i ; otherwise one term $1/p_i$ would already be at least $1/r$, leaving no room for the remaining positive terms in (9.26).

For $t > 0$, let

$$f_i(t) = \left(\frac{t^{x_i-1}}{e^t - 1} \right)^{1/p_i}. \quad (9.31)$$

Then

$$\left(\prod_{i=1}^m f_i(t) \right)^r = \frac{t^{r \sum_i (x_i - 1)/p_i}}{e^t - 1} = \frac{t^{T-1}}{e^t - 1}. \quad (9.32)$$

Consequently,

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r(0, \infty)}^r = \int_0^\infty \frac{t^{T-1}}{e^t - 1} dt = \Gamma(T) \zeta(T). \quad (9.33)$$

Generalized Holder gives

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r} \leq \prod_{i=1}^m \|f_i\|_{L^{p_i}}. \quad (9.34)$$

Moreover,

$$\|f_i\|_{L^{p_i}}^{p_i} = \int_0^\infty \frac{t^{x_i-1}}{e^t - 1} dt = \Gamma(x_i) \zeta(x_i). \quad (9.35)$$

Combining (9.33), (9.34), and (9.35), then raising to the power r , gives

$$\Gamma(T) \zeta(T) \leq \prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}. \quad (9.36)$$

Division by $\Gamma(T)$ proves (9.28). This solves the generalized Holder problem in Sroysang's notation, where his $\xi(s)$ is $\zeta(s)$ for $s > 1$ [2]. $\square$

Remark 9.4 (Free-energy interpretation of the Sroysang inequality). The same result follows from log-convexity of $M(s) = \Gamma(s) \zeta(s)$. With $\lambda_i = r/p_i$ and $T = \sum_i \lambda_i x_i$, Jensen's inequality for $\log M$ gives

$$M(T) \leq \prod_{i=1}^m M(x_i)^{\lambda_i} = \prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}, \quad (9.37)$$

which is equivalent to (9.28).

10 Laplace moment-ratio application

The complete-monotonicity proofs above repeatedly convert special-function curvature questions into positivity statements for Laplace kernels. The next lemma records a reusable moment-ratio consequence of that viewpoint.

Lemma 10.1 (Laplace moment-ratio bridge). *Let μ be a positive Borel measure on $[0, \infty)$, and assume that the tilted moments*

$$a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t) \quad (10.1)$$

are finite and positive for the displayed indices. Then, for every $n \geq 0$,

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x) a_{n+2}(x)} \quad (10.2)$$

is strictly increasing in x .

Proof. Put $r_j(x) = a_{j+1}(x)/a_j(x)$. Cauchy–Schwarz gives

$$a_{j+1}(x)^2 \leq a_j(x)a_{j+2}(x), \quad (10.3)$$

so $r_j(x) \leq r_{j+1}(x)$. Since $a'_j(x) = -a_{j+1}(x)$, differentiating (10.2) logarithmically gives

$$\frac{d}{dx} \log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x). \quad (10.4)$$

The right-hand side is at least $r_n(x) > 0$, because $r_{n+2}(x) \geq r_{n+1}(x)$. Hence B_n is strictly increasing. $\square$

Yin and Zhang define the Nielsen k -beta function by

$$\beta_k(x) = \int_0^1 \frac{t^{x-1}}{1+t^k} dt = \int_0^\infty \frac{e^{-xt}}{1+e^{-kt}} dt, \quad k > 0, x > 0. \quad (10.5)$$

They ask for the monotonicity of the derivative ratio involving $x\beta_k(x)$ [8]. The bridge lemma gives the exact parity law.

Theorem 10.2 (APP-0010: Nielsen k -beta derivative-ratio monotonicity). *Let $k > 0$, let $n \geq 0$, and put $f_k(x) = x\beta_k(x)$. Then*

$$R_{k,n}(x) = \frac{f_k^{(n+1)}(x)}{f_k^{(n)}(x)f_k^{(n+2)}(x)} \quad (10.6)$$

is strictly increasing on $(0, \infty)$ when n is odd, and strictly decreasing on $(0, \infty)$ when n is even.

Proof. From (10.5),

$$f_k(x) = x \int_0^\infty \frac{e^{-xt}}{1+e^{-kt}} dt. \quad (10.7)$$

Integrating by parts gives

$$f_k(x) = \frac{1}{2} + \int_0^\infty e^{-xt} \frac{ke^{-kt}}{(1+e^{-kt})^2} dt. \quad (10.8)$$

Thus f_k is a Laplace transform of a positive measure consisting of an atom $1/2$ at 0 and a positive density on $(0, \infty)$. Define

$$a_j(x) = (-1)^j f_k^{(j)}(x). \quad (10.9)$$

For the needed indices these are positive tilted moments of the measure in (10.8). Lemma 10.1 shows that

$$\frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)} \quad (10.10)$$

is strictly increasing. But

$$R_{k,n}(x) = (-1)^{n+1} \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}. \quad (10.11)$$

Multiplication by $(-1)^{n+1}$ gives strict increase for odd n and strict decrease for even n . This resolves the Nielsen k -beta derivative-ratio problem with the precise parity convention. $\square$

11 Endpoint and parameter applications

The next three applications add one certified endpoint result and two parameter-monotonicity results to the complete-monotonicity layer. Only fully solved source questions are labelled.

11.1 Exact $n = 2$ beta-window endpoint

For $s > 1$, write

$$Z_s(a) = \sum_{m=0}^{\infty} (m+a)^{-s}. \quad (11.1)$$

In Qi–Lim–Nantomah’s notation [4], put

$$C_2(x) = \psi''(x) + x\psi^{(3)}(x), \quad P_2(x) = \psi''(x)\psi''(1/x), \quad (11.2)$$

and define

$$\mathcal{I}_2 = \{\beta \in \mathbb{R} : x^\beta C_2(x) - P_2(x) < 0 \text{ for all } x > 0\}. \quad (11.3)$$

The remaining endpoint is controlled on $0 < x < 1$ by

$$R(x) = \frac{P_2(x)}{C_2(x)} = \frac{2Z_3(x)Z_3(1/x)}{3xZ_4(x) - Z_3(x)}, \quad Q_2(x) = \frac{\log R(x)}{\log x}. \quad (11.4)$$

Theorem 11.1 (APP-0011: exact $n = 2$ beta-window endpoint). *Let*

$$J = \left[\frac{287345}{1000000}, \frac{287346}{1000000} \right]. \quad (11.5)$$

There is a unique $\xi \in J$ at which Q_2 attains its global supremum on $(0, 1)$. Hence, with $L_2 = Q_2(\xi)$,

$$\mathcal{I}_2 = (L_2, 3]. \quad (11.6)$$

Proof. The identity $\psi^{(m)}(x) = (-1)^{m+1}m!Z_{m+1}(x)$ gives (11.4). Define

$$\Lambda(x) = -\frac{3Z_4(x)}{Z_3(x)} + \frac{3Z_4(1/x)}{x^2Z_3(1/x)} - \frac{6Z_4(x) - 12xZ_5(x)}{3xZ_4(x) - Z_3(x)} \quad (11.7)$$

and

$$G(x) = x \log x \Lambda(x) - \log R(x). \quad (11.8)$$

Differentiating (11.4) gives

$$Q_2'(x) = \frac{G(x)}{x(\log x)^2}. \quad (11.9)$$

All inequalities below are rational interval certificates using the Hurwitz-zeta integral-tail enclosure and the atanh logarithm enclosure. First,

$$G\left(\frac{287345}{1000000}\right) > 0, \quad G\left(\frac{287346}{1000000}\right) < 0. \quad (11.10)$$

Second, on

$$B = \left[\frac{1409}{5000}, \frac{293}{1000} \right],$$

the same enclosure proves

$$H(x) = \Lambda(x) + x\Lambda'(x) > 0. \quad (11.11)$$

Since $G'(x) = \log x H(x)$ and $\log x < 0$ on B , G is strictly decreasing on B . Together with (11.10), this gives a unique zero $\xi \in J$, and (11.9) shows that Q_2 is increasing before ξ and decreasing after ξ on B .

It remains only to exclude points outside the compact bracket. The endpoint certificate gives a rational lower witness $q_J < Q_2(\xi)$ with

$$\frac{115727}{50000} < q_J. \quad (11.12)$$

On $(0, 9/50]$ the scaled near-zero bound gives $Q_2(x) < 23/10 < q_J$. On $[9/10, 1)$, a finite rational subdivision proves $R(x) > 1$, hence $Q_2(x) < 0 < q_J$. On

$$\left[\frac{9}{50}, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, \frac{9}{10} \right],$$

a finite rational cover proves $Q_2(x) < q_J$: the left bridge uses

$$R(x) > Z_3(1/x) > x^{1157/500},$$

and the remaining intervals use the direct comparison

$$\log R(x) > \frac{115727}{50000} \log x.$$

Thus no point outside B reaches the value at ξ , so $L_2 = Q_2(\xi)$.

The source reduction gives the upper endpoint 3 and the convention that the lower endpoint is excluded when equality occurs. At $\beta = L_2$, equality occurs at $x = \xi$; therefore the exact admissible set is $(L_2, 3]$. $\square$

11.2 Baricz V_q parameter log-convexity

Baricz records the question whether, for fixed $x > 0$, the function

$$V_q(x) = \frac{2e^{x^2}}{\Gamma(q+1)} \int_x^\infty e^{-t^2} (t^2 - x^2)^q dt, \quad q > -1, \quad (11.13)$$

is log-convex as a function of q [9].

Theorem 11.2 (APP-0012: Baricz V_q strict q -log-convexity). *For each fixed $x > 0$, the map $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$.*

Proof. Put $u = t^2 - x^2$. Then

$$V_q(x) = \frac{1}{\Gamma(q+1)} \int_0^\infty e^{-u} u^q (u + x^2)^{-1/2} du. \quad (11.14)$$

Using

$$r^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-rs} ds$$

and Tonelli's theorem, we obtain

$$V_q(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} (1+s)^{-(q+1)} ds. \quad (11.15)$$

Let $a = q + 1 > 0$. Differentiation under the integral gives

$$(-1)^n \frac{d^n}{da^n} V_{a-1}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty (\log(1+s))^n s^{-1/2} e^{-x^2 s} (1+s)^{-a} ds \geq 0. \quad (11.16)$$

Equivalently, after the change of variables $y = \log(1+s)$, $V_{a-1}(x)$ is a positive Laplace transform in a . The representing measure is not concentrated at a single point, because the density in (11.15) is positive on $s > 0$. Strict Holder inequality for two different exponents therefore gives strict log-convexity in a , and the shift $a = q + 1$ gives the stated strict log-convexity in q . $\square$

11.3 Yin (p, k) -digamma sharp necessity

Yin proved sufficiency of $\alpha \leq 1$ for complete monotonicity of

$$\delta_{p,k,\alpha}(x) = x^\alpha \left[\frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \psi_{p,k}(x) \right], \quad p \in \mathbb{N}, \quad k > 0, \quad (11.17)$$

and asked whether this condition is also necessary [10].

Theorem 11.3 (APP-0013: Yin (p, k) -digamma sharp α -necessity). *If $\delta_{p,k,\alpha}$ is completely monotone on $(0, \infty)$, then $\alpha \leq 1$. Combined with Yin's sufficiency theorem, the sharp condition is $\alpha \leq 1$.*

Proof. The source representation is

$$\psi_{p,k}(x) = \frac{1}{k} \log(pk) - \sum_{n=0}^p \frac{1}{x + nk}. \quad (11.18)$$

Therefore the bracket in (11.17) is

$$B_{p,k}(x) = \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=0}^p \frac{1}{x + nk}. \quad (11.19)$$

With $t = x/k$,

$$B_{p,k}(x) = \frac{1}{k} \left[\sum_{n=0}^p \frac{1}{t + n} + \log \frac{t}{t + p + 1} \right].$$

Since $u \mapsto (t + u)^{-1}$ is strictly decreasing,

$$\sum_{n=0}^p \frac{1}{t + n} > \int_0^{p+1} \frac{du}{t + u} = \log \frac{t + p + 1}{t},$$

so $B_{p,k}(x) > 0$ for $x > 0$.

As $x \rightarrow 0^+$, (11.19) gives

$$B_{p,k}(x) = \frac{1}{x} + O(|\log x|), \quad \delta_{p,k,\alpha}(x) \sim x^{\alpha-1}. \quad (11.20)$$

If $\alpha > 1$, then $\delta_{p,k,\alpha}(x) \rightarrow 0$ as $x \rightarrow 0^+$. But a completely monotone nonzero function is positive and nonincreasing on $(0, \infty)$. For any fixed $x_0 > 0$, positivity gives $\delta_{p,k,\alpha}(x_0) > 0$, while the limit at 0 gives a point $0 < y < x_0$ with $\delta_{p,k,\alpha}(y) < \delta_{p,k,\alpha}(x_0)$, contradicting nonincreasingness. Hence $\alpha \leq 1$. $\square$

12 The Laplace-Transport Application Layer

The v009 applications are organized by one theme. A problem is first converted into a positive-kernel statement, usually by a logarithmic second difference, a Frullani identity, a moment determinant, or a transform normalization. The conclusion then follows by transporting that kernel through a permanence operation: Stieltjes transformation, Bernstein integration, stable subordination, quotient compression, or triangular coefficient extraction. This section records the ten new public solves.

Theorem 12.1 (APP-0014: Ramanujan integral is Stieltjes). *Let*

$$I_R(x) = \int_0^\infty \frac{e^{-xt}}{t(\pi^2 + \log^2 t)} dt, \quad x > 0.$$

Then I_R is a Stieltjes function. Moreover every primitive of the form

$$\tilde{I}_R(x) = a + \int_0^\infty \frac{1 - e^{-xt}}{t(\pi^2 + \log^2 t)} dt, \quad a \geq 0,$$

is a complete Bernstein function.

Proof. For $t > 0$,

$$\frac{1}{t(\pi^2 + \log^2 t)} = \frac{1}{\pi(1+t)} \int_0^1 t^{-u} \sin(\pi u) du.$$

The right side is a positive superposition of completely monotone powers and is therefore completely monotone as a function of t . The Laplace transform of a completely monotone density is Stieltjes. Integrating the same density against $1 - e^{-xt}$ gives a complete Bernstein function by the standard Stieltjes-density characterization of complete Bernstein functions. $\square$

Theorem 12.2 (APP-0015: Prabhakar Q-measure by stable subordination). *Let $0 < \alpha < 1$, $\eta > 0$, $\beta > \alpha\eta$, and $\lambda > 0$. Let*

$$E_{\alpha,\beta}^\eta(z) = \sum_{m=0}^\infty \frac{(\eta)_m z^m}{m! \Gamma(\alpha m + \beta)}$$

be the Prabhakar Mittag-Leffler function. Let $P_{\alpha,\beta}^\eta$ be the transform-normalized Pollard measure characterized in the strict Pollard range by

$$\int_0^\infty e^{-xr} dP_{\alpha,\beta}^\eta(r) = E_{\alpha,\beta}^\eta(-x), \quad x > 0.$$

Let S_α be the positive α -stable variable satisfying $\mathbb{E}e^{-sS_\alpha} = e^{-s^\alpha}$. Define the finite measure

$$Q_{\alpha,\beta}^\eta(A \mid \lambda) = \int \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\} dP_{\alpha,\beta}^\eta(r).$$

Then

$$\int_0^\infty e^{-xt} dQ_{\alpha,\beta}^\eta(t \mid \lambda) = E_{\alpha,\beta}^\eta(-\lambda x^\alpha).$$

Proof. Conditioning on r and using the stable transform gives

$$\mathbb{E}e^{-x(\lambda r)^{1/\alpha} S_\alpha} = e^{-\lambda r x^\alpha}.$$

Integration against $P_{\alpha,\beta}^\eta$ gives the Pollard representation of $E_{\alpha,\beta}^\eta(-\lambda x^\alpha)$. The total mass is finite, equal to the value at $x = 0$, and no normalization to a probability measure is needed for the transform identity. $\square$

Theorem 12.3 (APP-0016: Mills-ratio bound at every derivative level). *Let*

$$M_L(t) = \int_0^\infty u^L e^{-tu - u^2/2} du, \quad L = 0, 1, 2, \dots$$

Put $m_L(t) = M_{L+1}(t)/M_L(t)$ and

$$U_L(t) = \frac{\sqrt{t^2(t^2 + 4L + 7)^2 + 4(L + 1)(t^2 + 4L + 8)(t^2 + 4L + 6)} - t(t^2 + 4L + 7)}{2(t^2 + 4L + 8)}.$$

Then $m_L(t) < U_L(t)$ for every $L \geq 0$ and $t > 0$. With

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1},$$

and

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n,$$

the classical Mills ratio $r(t) = M_0(t)$ satisfies

$$r(t) > \frac{B_{L+1} - U_L B_L}{P_{L+1} + U_L P_L} \quad (L \geq 0 \text{ even}),$$

and

$$r(t) < \frac{U_L B_L - B_{L+1}}{P_{L+1} + U_L P_L} \quad (L \geq 1 \text{ odd}).$$

Proof. The density defining M_L is a positive moment density. Its order-four Hankel-minor consequence gives

$$M_{L+4}M_L - 4M_{L+3}M_{L+1} + 3M_{L+2}^2 > 0.$$

Using the recurrence $M_{n+1} = nM_{n-1} - tM_n$, this determinant inequality reduces to a quadratic inequality for m_L ; solving the admissible positive root gives $m_L < U_L$. Iterating the same recurrence expresses M_L in the form

$$M_L = (-1)^L (P_L r - B_L),$$

and substituting the level- L quotient bound gives the displayed alternating upper and lower bounds for r . $\square$

Theorem 12.4 (APP-0017: a gamma quotient need not be Bernstein). *The assertion that the gamma quotient*

$$x \mapsto \frac{\Gamma(x+a)\Gamma(x+b)}{\Gamma(x)\Gamma(x+a+b)}$$

is Bernstein for all positive a, b is false.

Proof. Take $a = 2$, $b = 3$, and set $y = x - 2$. The quotient becomes

$$g(y) = \frac{y(y+1)}{(y+3)(y+4)}.$$

A direct differentiation gives

$$g'''(y) = \frac{36(y^4 + 8y^3 + 12y^2 - 40y - 94)}{(y+3)^4(y+4)^4}.$$

Thus $g'''(1) < 0$. If g were Bernstein, then g' would be completely monotone and hence $g''' \geq 0$. The displayed value contradicts that necessary condition. $\square$

Theorem 12.5 (APP-0018: exact supremum for the Qi-Guo tau threshold). *For $s \in \mathbb{N}$ and $t > 0$, define*

$$\tau(s, t) = \frac{1}{s} \left[t - (t + s + 1) \left(\frac{t}{t+1} \right)^{s+1} \right].$$

Let $a_ > 0$ be the unique positive solution of*

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s \in \mathbb{N}, t > 0} \tau(s, t) = \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639 \dots$$

The supremum is not attained.

Proof. Writing $u = t/(t+1)$ gives a two-parameter expression on $0 < u < 1$. The Euler-Lagrange equations reduce the interior extremal sequence to the scaling limit $u = e^{-a/s}$. The limiting profile is maximized exactly when $e^a = 1 + a + a^2$, and the resulting value is $a/(1 + a + a^2)$. The source one-variable monotonicity estimates bound the finite- s expressions by this limit, while the scaling sequence approaches it. Hence the number is a supremum and is not attained at finite s, t . $\square$

Theorem 12.6 (APP-0019: exact Bernstein range for the W-symbol). *Let $0 < \alpha < 1$, $\beta \geq 0$, and*

$$\Phi_{\alpha, \beta}(s) = s^\alpha (1 + (1 - \alpha)s^{\alpha-1})^{-\beta}.$$

Then $\Phi_{\alpha, \beta}$ is a Bernstein function if and only if

$$0 \leq \beta \leq 1.$$

Proof. The non-Bernstein half $\beta > 1$ follows from the source asymptotic sign criterion. For $0 \leq \beta \leq 1$, put $a = 1 - \alpha$. Differentiation gives $\Phi'_{\alpha, \beta}(s) = H(s^a)$, where

$$H(y) = y^{\beta-1} (y + a)^{-\beta} \left(\alpha + \frac{a^2 \beta}{y + a} \right).$$

Each factor is completely monotone on $(0, \infty)$ for $0 \leq \beta \leq 1$, and s^a is a Bernstein function. Composition therefore makes $\Phi'_{\alpha, \beta}$ completely monotone, so $\Phi_{\alpha, \beta}$ is Bernstein. $\square$

Theorem 12.7 (APP-0020: Qi h_λ degree-four conjecture is false). *Let*

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \quad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\text{cm}} h_\lambda = \deg_x^{\text{cm}} (-h_\mu) = 4$ exactly in the stated parameter ranges $\lambda \leq 0$, $\mu \geq 4$, is false. Here $\deg_x^{\text{cm}} g$ denotes the source convention: the supremal power r for which $x^r g(x)$ is completely monotone on $(0, \infty)$.

Proof. Near zero,

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12} \right) \frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then $x^4 h_\lambda(x) = (-\lambda/12)x + O(x^2)$, whose derivative is positive near zero. If $\lambda = 0$, the coefficient $\pi^2/3 - 37/12$ is positive, again giving a positive derivative for the degree-four transform near zero. Complete monotonicity would require the first derivative to be non-positive. The same small- x test applied to $-x^4 h_\mu$ excludes the claimed μ -range. Hence the degree-four conjecture fails. $\square$

Theorem 12.8 (APP-0021: Szabo's cutoff is $\alpha_0 = 1$). For $0 < d < 1$, define

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

The exact cutoff for complete monotonicity of $y^\alpha H_d(y)$ is

$$\alpha_0 = 1.$$

Proof. The source theorem proves complete monotonicity for $\alpha \leq 1$. Conversely, as $y \downarrow 0$,

$$H_d(y) = \frac{1-d}{y} + O(1).$$

Hence, for $\alpha > 1$,

$$y^\alpha H_d(y) = (1-d)y^{\alpha-1} + O(y^\alpha),$$

whose derivative is positive near zero. A completely monotone function must be non-increasing, so no $\alpha > 1$ is admissible. $\square$

Theorem 12.9 (APP-0022: Chen-Choi Gurland ratio is log-completely monotone). Let

$$F(x) = \frac{\Gamma(1/x)\Gamma(3/x)}{\Gamma(2/x)^2}, \quad x > 0.$$

Then F is logarithmically completely monotone; equivalently,

$$(-1)^n (\log F)^{(n)}(x) > 0, \quad n = 0, 1, 2, \dots$$

Proof. Set

$$A(u) = 2 \log(u+2) - \log(u+1) - \log(u+3).$$

The identity

$$A(u) = \int_0^\infty e^{-ut} \frac{e^{-t}(1-e^{-t})^2}{t} dt$$

shows that A is strictly completely monotone. The Weierstrass product for Γ gives

$$\log F(x) = \log \frac{4}{3} + \sum_{m \geq 1} A(mx).$$

Termwise differentiation is justified on compact subintervals of $(0, \infty)$, and each differentiated summand has the required strict alternating sign. This proves logarithmic complete monotonicity. $\square$

Theorem 12.10 (APP-0023: Sokal lambda-derivatives have triangular compression). Let $n, k \in \mathbb{N}_0$, $\lambda > 0$, $x > 0$, and let f be a C^∞ function on $(0, \infty)$. In Sokal's generalized-Stieltjes application, the quantities below are the Hankel-type expressions associated with a generalized Stieltjes function of order λ . For $j \geq 0$, set

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x), \quad a = n + \lambda.$$

For $0 \leq r \leq k$, define

$$F_{n,r}^{[\lambda]}(x) = \sum_{j=0}^r \binom{r}{j} (a+j)_{r-j} G_j(x).$$

Then, for every integer $0 \leq \ell \leq k$,

$$\partial_{\lambda}^{\ell} F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-\ell} \frac{k!}{r!} [z^{k-r}] (-\log(1-z))^{\ell} F_{n,r}^{[\lambda]}(x).$$

For $\ell > k$, $\partial_{\lambda}^{\ell} F_{n,k}^{[\lambda]}(x) = 0$. The coefficients in this triangular compression are non-negative.

Proof. The exponential generating function is

$$\mathcal{F}(z) = (1-z)^{-a} \sum_{j \geq 0} G_j(x) \left(\frac{z}{1-z} \right)^j \frac{1}{j!}.$$

Differentiating ℓ times with respect to λ multiplies $\mathcal{F}$ by $(-\log(1-z))^{\ell}$. Extracting coefficients gives the displayed triangular formula. Since $-\log(1-z) = \sum_{m \geq 1} z^m/m$ has non-negative coefficients, all compression coefficients are non-negative. $\square$

13 Conclusion

Version v009 organizes the public theory through a Laplace-transport framework. The zeta-law entropy layer supplies the compression viewpoint, the kernel criteria identify complete monotonicity at the source, and the application layer shows that the same positivity can be transported through Stieltjes, Bernstein, stable-subordination, determinant, and triangular-transform constructions. The staged public ledger now contains twenty-three solved applications with uniform status *Solved*.

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